

Project Title:

Rainbow Food List: A Comprehensive Nutrition and Food Guide Mobile App/AI-Based Diet Suggestion

1. Executive Summary:

Rainbow Food List is a mobile application designed to provide detailed information on nutrition, food composition, and dietary advice. This user-friendly platform will allow consumers to easily access knowledge about the nutritional components of various foods, including proteins, carbohydrates, fats, fiber, vitamins, and minerals. The app will be entirely free to use, with a premium version offering additional personalized features such as AI-assisted diet plans and consultations with professional nutritionists. The aim of this project is to empower users to make informed dietary choices and address specific health concerns by providing tailored recommendations, especially for patients with chronic conditions such as heart disease, kidney issues, and diabetes.

2. Project Objectives:

- **Provide Comprehensive Nutritional Information:** The app will offer detailed data about various foods, including their nutritional content (protein, carbohydrates, fats, vitamins, minerals) to enable users to make well-informed food choices.
- **Address Specific Dietary Needs:** Custom food suggestions will be provided based on the user's health conditions, such as heart disease, kidney disease, or diabetes.
- **Alternative Food Recommendations:** For users unable to access certain foods (animal-based products), the app will suggest cost-effective plant-based alternatives that meet the same nutritional requirements.
- **Low-Cost, High-Nutrition Foods:** The app will highlight affordable foods that are nutrient-dense, helping users maintain a healthy diet within their budget.
- **Premium Services:** A paid version of the app will include AI-assisted diet charts

3. App Features:

- **Nutritional Information:** Detailed breakdown of nutrients, components, and minerals in various foods.
- **Free Access:** The basic app is free for everyone to use.
- **Premium Version:** AI and nutritionist-assisted diet chart creation.
- **Diet Recommendations:** Personalized food suggestions based on specific health conditions (kidney, heart, diabetes, etc.).
- **Alternative Food Suggestions:** Plant-based alternatives for those unable to consume animal products.
- **Food Composition Details:** Breakdown of protein, carbohydrates, fat, fiber, vitamins, etc.
- **Health Data Analysis:** Machine learning model to analyze user health data (age, weight, conditions).
- **User Profiles & History:** Database to store user inputs and recommendation history.
- **Personalized Feedback:** User interface for feedback and progress tracking.
- **Public Deployment:** AI-based system accessible to all users.

4. Target Audience:

The Rainbow Food List is designed for a broad audience including:

- Health-conscious individuals looking to improve their dietary habits.
- People with chronic conditions such as heart disease, diabetes, or kidney problems who require tailored dietary recommendations.
- Individuals with food restrictions who need alternative dietary options.
- Budget-conscious users seeking affordable but nutritious food options.

5. Project Deliverables:

Phase	Task/Deliverable	Required Experts	Timeline (Days)	Cost (Taka)
1. AI-Powered Recommendation Engine for Personalized Diet Suggestions			196	2,921,000 BDT
1.1 Data Preparation	Data collection, cleaning, preprocessing, and validation	Data Engineer, Domain Expert	45	495,000 BDT
1.2 Feature Engineering	Feature selection, creation, and validation	Data Scientist, Domain Expert	32	416,000 BDT
1.3 ML Model Building	Algorithm selection, training, and validation	ML Engineer, Data Scientist	43	645,000 BDT
1.4 ML Model Experiment, Track, and Select	Run experiments, track performance, and select the best model	ML Engineer, Data Scientist	31	465,000 BDT
1.5 ML Model Deployment and Testing	Deploy model, integrate with backend, testing, and validation	ML Engineer, DevOps, QA Engineer	45	900,000 BDT

2. User-Friendly Web/Mobile Application			151	1,768,500 BDT
2.1 Application Design	Design wireframes, UX/UI design, and validation	UX/UI Designer	30	195,000 BDT
2.2 Front-End Development	Build UI, implement forms, validation, and dynamic elements	Front-End Developer, QA Engineer	45	517,500 BDT
2.3 Back-End Development	Build server logic, authentication, and API integration	Back-End Developer, Database Admin	46	621,000 BDT
2.4 Integration with AI Model	Integrate AI model with backend and test	Back-End Developer, ML Engineer	30	435,000 BDT
3. Dashboard for Tracking Dietary Progress and Receiving Insights			137	1,628,500 BDT
3.1 Dashboard Design	Design dashboard with charts, graphs, and insights	UX/UI Designer, Data Analyst	31	387,500 BDT
3.2 Data Collection and Storage	Set up data collection pipelines and storage logic	Back-End Developer, Data Engineer	30	345,000 BDT

3.3	Dashboard Front-End Development	Develop dashboard interface, implement visualizations	Front-End Developer, Data Analyst	44	528,000 BDT
3.4	Dashboard Testing and Optimization	Test, optimize performance, and ensure compatibility	QA Engineer, Front-End Developer	32	368,000 BDT

6. Hardware & Engineering Requirements

❖ Server Infrastructure:

- Cloud Servers: Amazon Web Services (AWS) or Google Cloud for storing food databases and user data securely.
- Database: PostgreSQL database capable of handling large volumes of food and user data.
- AI Model Training: High-performance GPUs (NVIDIA Tesla or A100) for training machine learning models.
- Mobile Compatibility: Devices for testing across Android and iOS.

❖ Software & Engineering Tools:

- App Development Tools: Android Studio, Xcode for mobile app development.
- Frontend Tools: React, JavaScript frameworks.
- Backend Tools: Django or Flask, PostgreSQL, and machine learning libraries like TensorFlow or PyTorch.
- AI Model Training Software: TensorFlow, PyTorch, Scikit-learn for building and refining AI models.

7. Summary of Budget Breakdown:

Phase 1: AI-Powered Recommendation Engine for Personalized Diet Suggestions:

- **Total Duration:** 196 days
- **Experts Needed:** Data Engineer, Domain Expert, Data Scientist, ML Engineer, DevOps, QA Engineer

1.1 Data Preparation (45 days)

- Required Experts: Data Engineer, Domain Expert
- Budget Allocation:
 - Data Engineer (45 days BDT 5,000/day): 225,000
 - Domain Expert (45 days BDT 6,000/day): 270,000
- **Total for Data Preparation: 495,000 BDT**

1.2 Feature Engineering (32 days)

- Required Experts: Data Scientist, Domain Expert
- Budget Allocation:
 - Data Scientist (32 days BDT 7,000/day): 224,000
 - Domain Expert (32 days BDT 6,000/day): 192,000
- **Total for Feature Engineering: 416,000 BDT**

1.3 ML Model Building (43 days)

- Required Experts: ML Engineer, Data Scientist
- Budget Allocation:
 - ML Engineer (43 days BDT 8,000/day): 344,000
 - Data Scientist (43 days BDT 7,000/day): 301,000
- **Total for ML Model Building: 645,000 BDT**

1.4 ML Model Experimentation, Tracking, and Selection (31 days)

- Required Experts: ML Engineer, Data Scientist
- Budget Allocation:
 - ML Engineer (31 days BDT 8,000/day): 248,000
 - Data Scientist (31 days BDT 7,000/day): 217,000
- **Total for Model Experimentation and Selection: 465,000 BDT**

1.5 ML Model Deployment and Testing (45 days)

- Required Experts: ML Engineer, DevOps, QA Engineer
- Budget Allocation:
 - ML Engineer (45 days BDT 8,000/day): 360,000
 - DevOps (45 days BDT 6,500/day): 292,500
 - QA Engineer (45 days BDT 5,500/day): 247,500
- **Total for Model Deployment and Testing: 900,000 BDT**

❖ **Total Budget for Phase 1: AI-Powered Recommendation Engine = 2,921,000 BDT**

Phase 2: User-Friendly Web/Mobile Application:

- **Total Duration: 151 days**
- **Experts Needed:** UX/UI Designer, Front-End Developer, Back-End Developer, Database Admin, QA Engineer, ML Engineer

2.1 Application Design (30 days)

- Required Experts: UX/UI Designer
- Budget Allocation:
 - UX/UI Designer (30 days BDT 6,500/day): 195,000
- **Total for Application Design: 195,000 BDT**

2.2 Front-End Development (45 days)

- Required Experts: Front-End Developer, QA Engineer
- Budget Allocation:
 - Front-End Developer (45 days BDT 6,000/day): 270,000
 - QA Engineer (45 days BDT 5,500/day): 247,500
- **Total for Front-End Development: 517,500 BDT**

2.3 Back-End Development (46 days)

- Required Experts: Back-End Developer, Database Admin
- Budget Allocation:
 - Back-End Developer (46 days BDT 6,500/day): 299,000
 - Database Admin (46 days BDT 7,000/day): 322,000
- **Total for Back-End Development: 621,000 BDT**

2.4 Integration with AI Model (30 days)

- Required Experts: Back-End Developer, ML Engineer
- Budget Allocation:
 - Back-End Developer (30 days BDT 6,500/day): 195,000
 - ML Engineer (30 days BDT 8,000/day): 240,000
- **Total for Integration with AI Model: 435,000 BDT**

❖ **Total Budget for Phase 2: User-Friendly Web/Mobile Application = 1,768,500 BDT**

Phase 3: Dashboard for Tracking Dietary Progress and Receiving Insights:

- **Total Duration: 137 days**
- **Experts Needed:** UX/UI Designer, Data Analyst, Back-End Developer, Data Engineer, QA Engineer, Front-End Developer

3.1 Dashboard Design (31 days)

- Required Experts: UX/UI Designer, Data Analyst
- Budget Allocation:
 - UX/UI Designer (31 days BDT 6,500/day): 201,500
 - Data Analyst (31 days BDT 6,000/day): 186,000
- **Total for Dashboard Design: 387,500 BDT**

3.2 Data Collection and Storage (30 days)

- Required Experts: Back-End Developer, Data Engineer
- Budget Allocation:
 - Back-End Developer (30 days BDT 6,500/day): 195,000
 - Data Engineer (30 days BDT 5,000/day): 150,000
- **Total for Data Collection and Storage: 345,000 BDT**

3.3 Dashboard Front-End Development (44 days)

- Required Experts: Front-End Developer, Data Analyst
- Budget Allocation:
 - Front-End Developer (44 days BDT 6,000/day): 264,000
 - Data Analyst (44 days BDT 6,000/day): 264,000
- **Total for Dashboard Front-End Development: 528,000 BDT**

3.4 Dashboard Testing and Optimization (32 days)

- Required Experts: QA Engineer, Front-End Developer
- Budget Allocation:
 - QA Engineer (32 days BDT 5,500/day): 176,000
 - Front-End Developer (32 days BDT 6,000/day): 192,000
- **Total for Dashboard Testing and Optimization: 368,000 BDT**
- Total Budget for Phase 3: Dashboard for Tracking Dietary Progress and Receiving Insights = 1,628,500 BDT

Summary of Total Budget in BDT for the AI Model and Application:

Expense Category	Estimated Cost
AI-Powered Recommendation Engine	2,921,000 BDT
User-Friendly Web/Mobile Application	1,768,500 BDT
Dashboard for Tracking Dietary Progress and Receiving Insights	1,628,500 BDT
Total Estimated Cost	6,318,000 BDT

8. Project Team:

Role	Responsibilities	Accountability	Consultancy	Informed	Remarks (Name of team member)
Project Manager / Product Owner	Project oversight, timely delivery, goal alignment, stakeholder communication, team coordination, Monitor and Evaluate	Project Manager	Core Development Team, Nutrition & Health Experts	All Team Members, Sponsors, Investors	(Debt)
Backend Developer (Python & PostgreSQL)	Backend architecture, database management, Python coding, data integration	Backend Developer	Project Manager, Data Engineer, ML Engineer	Mobile App Developer, Frontend Developer	(Debt)

Mobile App Developer (Android & iOS)	Mobile app design, UI/UX integration, performance optimization	Mobile App Developer	Project Manager, UI/UX Designer	Backend Developer, Frontend Developer	(Debt)
Frontend Developer (Web Interface)	Web interface development for backend access via browsers, UI element integration	Frontend Developer	Project Manager, UI/UX Designer	Mobile App Developer, Backend Developer	(Debt)
Machine Learning Engineer	AI solutions implementation, personalized diet recommendations, model design for nutrition predictions	Machine Learning Engineer	Project Manager, Data Scientist, Backend Developer	UI/UX Designer, App Developers	(Debt)
UI/UX Designer	Interface design for mobile/web platform, user-friendly and visually appealing layouts	UI/UX Designer	Project Manager, Mobile App Developer, Frontend Developer	All Developers	(Debt)
Food and Health Researchers	Food composition research, nutrient details, alternative foods, database maintenance	Food and Health Researchers	Nutrition Content Specialist, ML Engineer	Backend Developer, Nutrition Team	(Debt)
Marketing Manager	Marketing strategy, branding, social media, and PR targeting free and premium users	Marketing Manager	Project Manager, Content Writers	All Team Members, Sponsors, Investors	(Debt)
Content Writers / Nutrition Specialist	Nutrition content creation, food descriptions, SEO optimization	Content Writers	Project Manager, Food and Health Researchers	Marketing Team, App Developers	(Debt)

Customer Support Team	User queries, support, feedback, navigation assistance	Customer Support Team	Project Manager	Marketing Manager, App Developers	(Debt)
Financial Advisor / Accountant	Budget management, investor relations, financial forecasting, premium monetization strategy	Financial Advisor	Project Manager, Legal Advisor	Sponsors, Investors	(Debt)
Legal Advisor	Licensing, privacy policies, user agreements, regulatory compliance	Legal Advisor	Project Manager, Financial Advisor	Stakeholders, Sponsors, Investors	(Debt)
Founders / Owners	App vision, strategic direction, mission alignment, affordability, accessibility	Founders / Owners	Project Manager, Marketing Manager	Sponsors, Investors, All Team Members	(Daffodil International University)
Investors	Funding provision for development, marketing, scaling	Investors	Project Manager, Financial Advisor	Founders / Owners, Sponsors	(Daffodil International University)
Sponsors	Partnership support, event sponsorship, app content support	Sponsors	Project Manager, Marketing Manager	Founders / Owners, Investors	(Daffodil Family, Health Org.)

9. Project Risks and Mitigations

9.1 Team Member Turnover:

- **Risk:** A key team member leaves the project.
- **Mitigation:** Create a detailed documentation system and cross-train team members to share knowledge.

9.2 Budget Overruns:

- **Risk:** Development costs exceed budget.

- **Mitigation:** Set a strict budget limit and regularly review expenses; use agile methodology for incremental spending.

9.3 Data Privacy Issues:

- **Risk:** Breaches in handling user data.
- **Mitigation:** Implement strong security measures and compliance with data protection regulations.

9.4 Technical Challenges:

- **Risk:** Difficulties in integrating machine learning models or app features.
- **Mitigation:** Conduct thorough research before implementation and consider hiring specialized consultants if necessary.

9.5 Market Competition:

- **Risk:** Existing apps may overshadow the new app.
- **Mitigation:** Conduct market analysis and focus on unique features or partnerships that differentiate the app.

9.6 User Acceptance:

- **Risk:** Low user adoption rate post-launch.
- **Mitigation:** Engage potential users in beta testing and gather feedback to improve usability before the full launch.

9.7 Compliance and Regulatory Issues:

- **Risk:** Failure to meet health and nutritional guidelines.
- **Mitigation:** Consult with legal experts to ensure compliance with relevant regulations.

10. Project Pilot:

- **Run the test project.**
- **Date: 05/05/2026**

11. Maintenance Plan

11.1. Continuous Monitoring of System Performance

- Objective: Ensure optimal app performance and user experience.
- Actions:
 - Implement performance monitoring tools (New Relic, Google Analytics) to track app usage and system health.
 - Set up alerts for critical performance metrics (response time, downtime).

- Conduct regular system audits to identify bottlenecks or performance issues.

11.2. Monthly Updates to Improve AI Algorithms Based on User Feedback

- Objective: Enhance AI capabilities and user satisfaction.
- Actions:
 - Collect user feedback through surveys and in-app feedback mechanisms.
 - Analyze feedback to identify trends and areas for improvement in AI recommendations.
 - Schedule monthly updates to algorithms based on insights gathered, incorporating the latest nutritional research.

11.3. Regular Bug Fixes, Security Updates, and User Support

- Objective: Maintain app stability and security while providing support to users.
- Actions:
 - Establish a bug tracking system (Jira, GitHub Issues) to monitor and prioritize bug fixes.
 - Schedule regular security audits and apply patches to address vulnerabilities.
 - Provide a support channel (chat, email) for user inquiries and issues, ensuring timely responses.

11.4 Ongoing User Training and Documentation Updates

11.4.1. Documentation and Training Roles and Responsibilities

- **Project Manager / Product Owner:** Oversees the user training strategy and documentation quality, ensuring timely updates and alignment with project goals.
- **Content Writers / Nutrition Specialist:** Creates clear, user-centric documentation, guides, and nutrition content, optimized for accessibility and SEO.
- **UI/UX Designer:** Designs visually engaging training materials, such as tutorials, within the app or as downloadable resources, ensuring ease of navigation.
- **Customer Support Team:** Manages user training queries, feedback collection, and documentation of FAQs to inform updates and future training sessions.

- **Backend Developer:** Supports integration of training modules within the backend, ensuring user data privacy.
- **Marketing Manager:** Manages communication of training updates to users via newsletters, social media, and in-app notifications.

11.4.2. Training Format and Materials

- **Interactive Tutorials and Walkthroughs:** Embedded within the app for onboarding, new feature updates, and troubleshooting.
- **Video Guides and Infographics:** Designed by the UI/UX Designer and Content Writer for step-by-step visual guidance, hosted on the app's support page.
- **FAQs and Help Articles:** Regularly updated based on user feedback, covering common user queries, app usage, and nutrition information.
- **In-App Notifications and Push Alerts:** Managed by the Marketing Manager, these alerts inform users of updates, new features, and training resources.
- **Newsletter Content:** Monthly digest with health tips, usage insights, and app updates, crafted by the Content Writers and sent to users by the Marketing Team.

11.4.3. Update Frequency and Accountability

- **Monthly Updates:** Minor documentation updates for bug fixes, new recipes, and any user-facing feature adjustments. **Accountability:** Project Manager, Content Writers.
- **Quarterly Training Sessions:** For major feature launches or significant updates; materials are prepared in advance by the UI/UX Designer and Content Writers. **Accountability:** Marketing Manager, Customer Support Team.
- **Annual Comprehensive Review:** A complete audit of user documentation, incorporating lessons learned from user feedback, support tickets, and feature improvements. **Accountability:** Project Manager, Customer Support, Content Writers.

11.4.4. Feedback Collection and Continuous Improvement

- **User Feedback Forms:** In-app feedback prompts and surveys after updates.
- **Customer Support Analysis:** Regular analysis of common support queries, handled by the Customer Support Team, to identify documentation gaps.

- **User Testing:** Focus groups for major training material or documentation revamps, coordinated by the Project Manager and UI/UX Designer.

11.4.5. Tools and Resources

- **Knowledge Base Software:** Centralized documentation tool accessible for users.
- **Analytics and Tracking:** Monitor user engagement with training materials and documentation using in-app metrics to identify which resources are most used or where users drop off.
- **Cross-Platform Accessibility:** Ensure resources are optimized for both mobile and web formats to reach the app's diverse user base.

11.4.6. Reporting and Informed Stakeholders

- **Monthly Report:** Summarizing training material effectiveness and user satisfaction, shared with the Founders, Investors, and key stakeholders.
- **Quarterly Stakeholder Briefs:** Updates on training progress, feedback integration, and upcoming improvements.

12. Project Handover:

12.1. Handover Preparation

- Documentation Collection and Organization:
 - Ensure all documents related to the project are gathered and organized. This includes:
 - Functional and technical specifications
 - User manuals and guides
 - System architecture and infrastructure diagrams
 - Database schemas
 - API documentation
 - Machine learning model documentation (if any)
 - Test cases and results
 - Deployment scripts and installation guides
 - Version control documentation
 - Development environment setup instructions

- **Final Testing and Sign-off:**
 - Ensure the app has passed all rounds of testing, including functional, performance, and security testing.
 - Obtain the sign-off from the QA and project management teams before the official handover.

12.2. Knowledge Transfer Sessions

- **Handover Training:**
 - Conduct comprehensive training sessions for the operations and support teams. Topics covered in the training include:
 - **App Functionality:** Walkthrough of all app features (both free and premium versions).
 - **Backend System:** Detailed explanation of the Python web application, PostgreSQL database, and machine learning components.
 - **Error Handling:** Overview of error codes, logs, and troubleshooting procedures.
 - **App Deployment:** Guide the team through the process of deploying updates and patches using the pre-determined deployment pipelines.
 - **System Maintenance:** Instructions on how to perform regular system maintenance tasks (backups, server monitoring, etc.).
 - **Monitoring Tools:** Overview of the monitoring systems in place to ensure continuous uptime and performance.

12.3. Transfer of Responsibility

- **Ownership Transfer:**

Assign and document the roles and responsibilities of the operations and support teams for the ongoing maintenance and support of the system.

- **Support Team Responsibilities:**
 - Manage day-to-day user support and incident response.
 - Handle premium customer service, including diet chart queries.
 - Update the app with the latest nutrition and food data regularly.
- **Operations Team Responsibilities:**
 - Manage server infrastructure (Python, PostgreSQL) and monitor system performance.
 - Perform regular security audits, database maintenance, and backups.

- Handle scaling issues as the app grows in user base.
- Ensure that machine learning models are retrained or updated as needed.

12.4. System Monitoring and Support Plan

- **Maintenance Plan:**

- Establish a schedule for routine system checks (database integrity, server health, app performance).
- Ensure the app's uptime and performance are tracked using monitoring tools.
- Perform regular updates for app security and optimization.

- **Escalation Procedures:**

- Define the protocol for escalating issues that cannot be resolved by the support team.
- Ensure critical incidents are addressed within pre-defined Service Level Agreements (SLAs).

12.5. Post-Deployment Review & Updates

- **Feedback Loop:**

- Collect feedback from the support and operations teams after deployment.
- Perform a post-deployment review to identify any remaining bugs, issues, or enhancement opportunities.

- **Authority Updates:**

- Provide regular status updates to project stakeholders and the authority.
- Inform them of any updates or changes to the system post-launch, such as new features, bug fixes, or important incidents.
- Share insights from the feedback gathered, highlighting the app's performance, user adoption, and areas for improvement.

12.6. Final Sign-Off

- **Handover Completion:**

- Obtain formal acknowledgment from the support and operations teams that they are fully trained and equipped to manage the "Rainbow Food List" app.
- Ensure the authority is informed and approves the transition to ongoing operations.

13. Communication Plan:

13.1. Weekly Team Meetings

Audience: Internal project team (developers, designers, data scientists, nutritionists)

- **Purpose:** Share progress updates, review potential risks, and discuss upcoming deliverables.
- **Frequency:** Weekly
- **Format:** Online meeting (video conferencing)
- **Key Points:**
 - Project milestones progress
 - Current tasks and deliverables
 - Risk assessment and mitigation
 - Discussion of technical issues or development challenges

13.2. Bi-monthly Sponsor Meetings

- **Audience:** Project sponsors, key stakeholders (investors, management)
- **Purpose:** Provide high-level updates, review the budget and project status.
- **Frequency:** Bi-monthly
- **Format:** Virtual/in-person presentation
- **Key Points:**
 - Overall project health and milestones
 - Financial health (budget and resource utilization)
 - Feedback and alignment on strategic decisions
 - Next-phase planning

13.3. User Communication (Pilot Phase)

- **Audience:** Selected pilot users
- **Purpose:** Share regular updates, gather feedback, and resolve issues.
- **Frequency:** Weekly or bi-weekly (depending on pilot duration)
- **Format:** Emails, surveys, and in-app notifications
- **Key Points:**
 - Key features and instructions for usage
 - Feedback collection on usability, functionality, and design
 - Issue resolution support channels

13.4. Post-Deployment Communication

- **Audience:** All users of the mobile app
- **Purpose:** Continuous engagement with users through updates, support, and feedback.
- **Frequency:** Ongoing (monthly newsletters, ad-hoc support updates)
- **Format:** Newsletters, in-app notifications, surveys, and support tickets
- **Key Points:**
 - Product updates and new feature announcements
 - Tips on app usage, personalized content, or nutrition advice
 - User satisfaction surveys for further app improvements
 - Dedicated support channels for resolving issues

14. EVM Project Measurement:

14.1. Key Performance Indicators (KPIs)

- **Planned Value (PV):** The budgeted cost for the work scheduled up to a specific date.
- **Earned Value (EV):** The budgeted cost for the work actually completed up to that date.
- **Actual Cost (AC):** The actual cost incurred for the work completed up to that date.

14.2 EVM Formulae

- **Cost Performance Index (CPI):**
$$CPI = \frac{EV}{AC}$$
- **Schedule Performance Index (SPI):**
$$SPI = \frac{EV}{PV}$$

14.3. Phase 1: AI-Powered Recommendation Engine for Personalized Diet Suggestions

❖ Summary

- **Duration:** 196 days
- **Budget:** 2,921,000 BDT

❖ Task Breakdown and EVM Tracking

Task	Duration (Days)	Planned Cost (BDT)	Actual Cost (BDT)	Earned Value (BDT)
1.1 Data Preparation	45	495,000		
1.2 Feature Engineering	32	416,000		
1.3 ML Model Building	43	645,000		
1.4 ML Model Experimentation	31	465,000		
1.5 ML Model Deployment and Testing	45	900,000		
Total	196	2,921,000		

❖ Performance Analysis

- **PV (Planned Value):** Calculated based on the schedule and budgeted cost for tasks completed.
- **EV (Earned Value):** Based on the completion status of the tasks.
- **AC (Actual Cost):** To be filled as the project progresses.

14.4. Phase 2: User-Friendly Web/Mobile Application

Summary

- **Duration:** 151 days
- **Budget:** 1,768,500 BDT

Task Breakdown and EVM Tracking

Task	Duration (Days)	Planned Cost (BDT)	Actual Cost (BDT)	Earned Value (BDT)
2.1 Application Design	30	195,000		
2.2 Front-End Development	45	517,500		
2.3 Back-End Development	46	621,000		
2.4 Integration with AI Model	30	435,000		
Total	151	1,768,500		

Performance Analysis

- **PV (Planned Value):** Calculated based on the schedule and budgeted cost for tasks completed.
- **EV (Earned Value):** Based on the completion status of the tasks.
- **AC (Actual Cost):** To be filled as the project progresses.

14.5. Phase 3: Dashboard for Tracking Dietary Progress and Receiving Insights

❖ Summary

- **Duration:** 137 days
- **Budget:** 1,628,500 BDT

❖ Task Breakdown and EVM Tracking

Task	Duration (Days)	Planned Cost (BDT)	Actual Cost (BDT)	Earned Value (BDT)
3.1 Dashboard Design	31	387,500		
3.2 Data Collection and Storage	30	345,000		
3.3 Dashboard Front-End Development	44	528,000		
3.4 Dashboard Testing and Optimization	32	368,000		
Total	137	1,628,500		

Performance Analysis

- **PV (Planned Value):** Calculated based on the schedule and budgeted cost for tasks completed.
- **EV (Earned Value):** Based on the completion status of the tasks.
- **AC (Actual Cost):** To be filled as the project progresses.

Overall Project Performance Metric

Combined Metrics

- **Total Planned Value (PV):** 6,318,000 BDT
- **Total Earned Value (EV):** (To be calculated based on completed tasks)
- **Total Actual Cost (AC):** (To be calculated as work is completed)

Indices and Forecasting

- **CPI and SPI** will be calculated as data is entered for EV and AC.
- **Forecasting:** Use the EVM metrics to forecast future performance, making adjustments to maintain project schedules and budgets.

Thank You.